Case Study

Vertical Dynamics Analysis of a Passenger Van

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Abstract

This experimental study aims to characterize the vertical dynamics of a road vehicle and validate the theoretical two-degree-of-freedom quarter-car model. Testing is conducted on a Mercedes Vito passenger van equipped with accelerometers on both the sprung mass (chassis) and unsprung mass (hub carrier). Data is acquired across three distinct driving scenarios: traversing a wood plank, driving an undefined urban route, and completing a closed-loop urban circuit. Subsequent frequency spectrum and modal analyses in Dewesoft X identify the system's fundamental dynamic properties. The results consistently reveal a primary sprung mass heave mode at approximately 1.5 Hz to 1.6 Hz with a damping ratio between 25% and 34%. Additionally, a second mode corresponding to the unsprung mass oscillation is identified near 7 Hz with a damping ratio of 6.4%. Despite challenges such as a low signal-to-noise ratio from the sprung mass accelerometer and the hardware malfunction of the NAVION inertial platform, the empirical data successfully demonstrates the suspension system's behavior as a low-pass filter.

Contents

1 Introduction

The goal of this experimental activity is to characterize the vertical dynamics of a road vehicle. This is performed by placing accelerometers in strategic positions in order to obtain valuable results that can be compared with the "quarter car model" described in university courses.

The location set for this experiment is Dewesoft headquarters in Vitinia (Rome) and the vehicle used is a Mercedes Vito passenger van (Figure (1.0.1)).

Figure 1.0.1

1.1 Theoretical background: quarter car model

The quarter car model is a 2 degree of freedom (DOF) model used to describe the vertical dynamics of a vehicle (Figure (1.1.1))

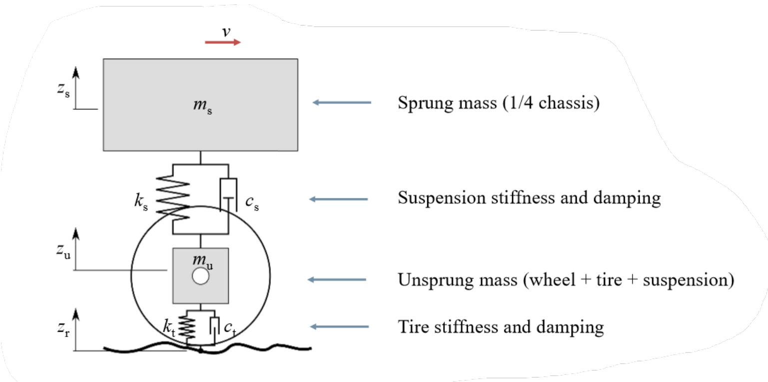

Figure 1.1.1

- $z_s \rightarrow$ vertical displacement of the sprung mass
- $z_u \rightarrow$ vertical displacement of the unsprung mass
- $z_r \rightarrow$ road irregularity profile (imposed displacement)

The equations of motion of the system are the following:

$$\begin{cases} m_s\ddot{z}_s + c_s(\dot{z}_s - \dot{z}_u) + k_s(z_s - z_u) = 0 \\ m_u\ddot{z}_u - c_s(\dot{z}_s - \dot{z}_u) - k_s(z_s - z_u) + c_t(\dot{z}_u - \dot{z}_r) + k_t(z_u - z_r) = 0 \end{cases} \tag{1.1}$$

It is important to clarify a few features/limitations of the model:

- k_s, c_s represent an equivalent stiffness and damping of the vehicle suspension, in the simplest formulation of the model they are assumed to be constant and the force-displacement or force-velocity relationship is assumed to be linear
- k_t, c_t represent an equivalent stiffness and damping of the tire, in the simplest formulation of the model they are assumed to be constant and the force-displacement or force-velocity relationship is assumed to be linear

2 Measurement setup

This section covers the setup of the experiment, highlighting the instrument selection, their positioning and the planning of the actual experimental activity.

2.1 Instruments used

The expected natural frequencies concerning the vertical motion of a road vehicle are expected to be in the range $0\,Hz \div 4\,Hz$ for the first mode and between $10\,Hz \div 20\,Hz$ for the second mode. Considering the experiment is performed on a van instead of a regular road vehicle the results might vary a bit but the order of magnitude is clearly not expected to change. Therefore, since the frequency range of interest is very low, it is important to verify that the accelerometers used are suitable for this type of measurement.
Two accelerometers are used: one is mounted on the unsprung mass while the other one on the sprung mass. In both cases, they are oriented in such a way that the positive vertical axis is pointing downwards:

- a triaxial accelerometer Bruel & Kjaer 4528 B is mounted on the hub carrier (in the **red circle** in Figure (2.1.1a)), the technical sheet[1] reports a lower frequency bound of $0.3\,Hz$
- a monoaxial accelerometer DYTRAN 3055D1T is mounted on the chassis (in the **blue circle** in Figure (2.1.1a)), the technical sheet[2] reports a lower frequency bound of $1\,Hz$, which may be more critical

A few remarks on this step:

- both accelerometers are glued in position. While this is not an issue for the triaxial accelerometer, it might be for the monoaxial one, which is designed to be stud mounted on a base; however, this is the only option with the available instrumentation
- the cables need to allow for wheel steering and vertical motion without entangling or getting caught in the moving parts, for this purpose, high-performance scotch tape is used
- the accelerometers are mounted on the front left wheel, which is close to the engine but it is not a driving wheel since the vehicle is RWD

[1] link to technical sheet of triaxial accelerometer
[2] link to technical sheet of monoaxial accelerometer

(a) (b)

Figure 2.1.1

In addition, a NAVION inertial navigation system (INS) platform is also used to track the vehicle's position, speed and pitch/roll angles; it is mounted on the roof with magnetic supports (Figure (2.1.2)). Note that for the purpose of this experiment the NAVION is not intended to serve as measurement device for the sprung mass acceleration because (compared to the sprung mass accelerometer), the signal is even more filtered by the dynamics of the chassis and the panels of the body.

Figure 2.1.2

The accelerometers are connected to a SIRIUS-HD-16xACC DAQ that is secured on the dashboard of the van. The whole acquisition setup is reported in Figures (2.1.3a), (2.1.3b)

(a)

(b)

Figure 2.1.3

2.2 Experimental plan

The measurements are performed in three different scenarios:

1. driving over a wood plank in the parking lot (Figure (2.2.1))
    - advantages (✓):
        - good repeatability
    - disadvantages (×):

– low speed due to limited available space

Figure 2.2.1

2. driving on urban road (without a specific path to follow) (Figure (2.2.2))
    - advantages (✓):
        – large variety of data is acquired (speed bumps, potholes, road irregularity)
    - disadvantages (×):
        – low repeatability

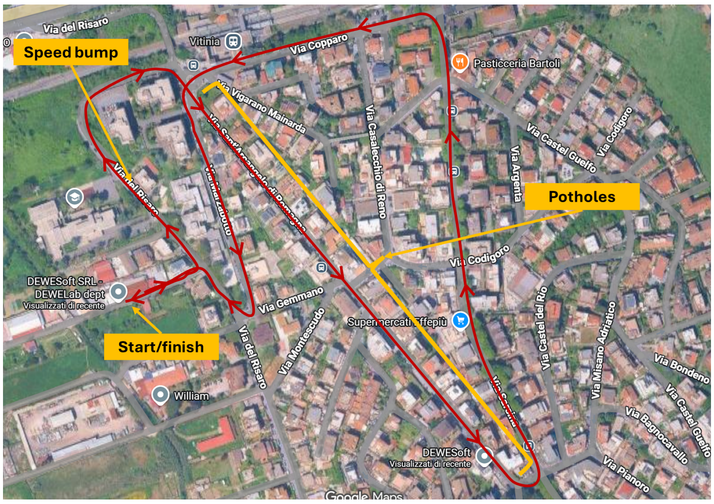

Figure 2.2.2

3. driving on urban road (following a well defined path) (Figure (2.2.3))

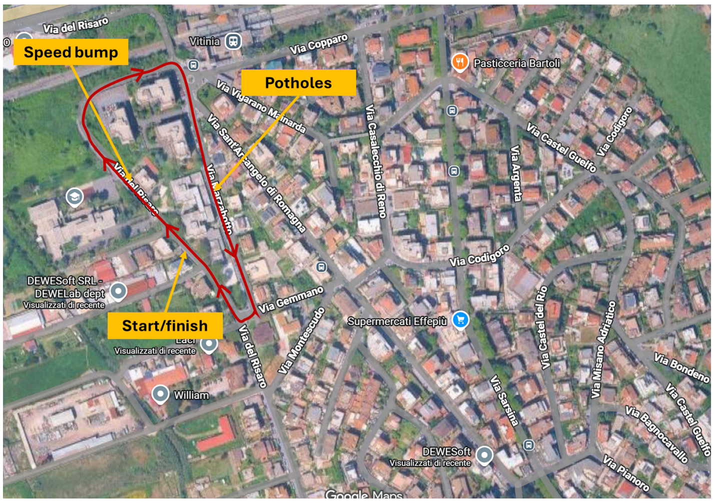

Figure 2.2.3

- three laps are covered around this closed loop
- this could potentially be a compromise between the two other scenarios

3 Analysis of the results

This section covers the analysis of the results obtained in the three different scenarios described previously. At first a simple frequency spectrum analysis is performed; subsequently, the transfer function between the unsprung and sprung mass acceleration is computed relying on the modal test and modal analysis modules. Note that the system (the vehicle) is always the same, consequently, the same results are expected to be found; actually, this can be a validation criterion to assess their consistency.

The setup used for the three analyses is generally the same and it is described in section (3.1), minor changes will be highlighted subsequently.

3.1 Dewesoft X setup

3.1.1 FFT math setup

- window function: Hanning, 50% overlap
- frequency resolution: $0.5\,Hz$ $\rightarrow$ the two natural frequencies of the system (described in Section 2.1) are expected to be well separated. The natural frequencies should be identified even if the frequency resolution is "coarse"
- averaging mode: linear, overall
- DC cutoff frequency: $0.1\,Hz$

3.1.2 Modal test module setup

- test method: ODS
- excitation:
  - FFT window: Hanning, 50% overlap
  - input: unsprung mass acceleration
- response:
  - input: sprung mass acceleration
- FRF:
  - FRF frequency resolution: $0.5\,Hz$
  - averaging mode: linear, unlimited averages
- outputs:

– FRF H1 estimator is used (more noise is observed on the output than on the input). For the sake of completeness, the H2 is computed but it is not used because poor results are obtained

3.1.3 Modal analysis setup

- max order: 24 (default setting)
- min frequency: $0\,Hz$
- max frequency: $50\,Hz$

3.2 Driving over wood plank

The recordings of the different events have been merged sequentially in a single time history:

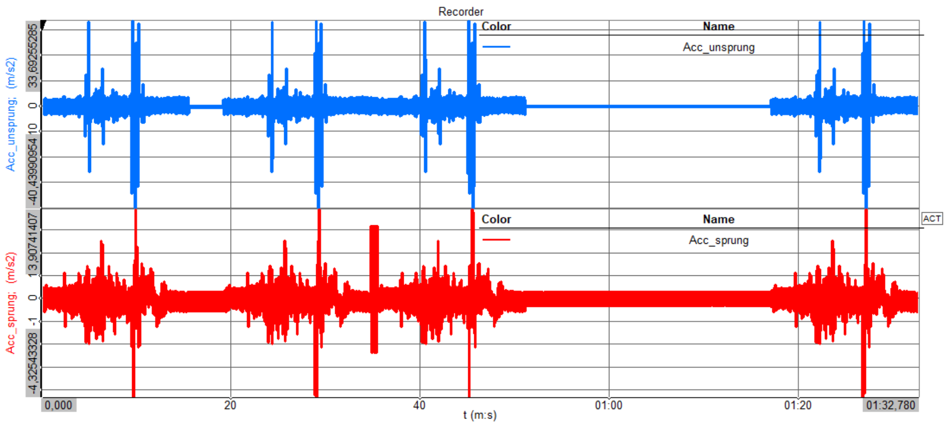

Figure 3.2.1

- four passages over the wood plank are reported in the plot. In each passage, the first spike corresponds to the wheel hitting the plank, while the second spike corresponds to the wheel jumping off
- a major issue can be seen immediately: for the sprung mass accelerometer the signal-to-noise ratio is low. This is inevitably caused by the non optimal mounting position of the instrument, that happens to be too close to the engine compartment

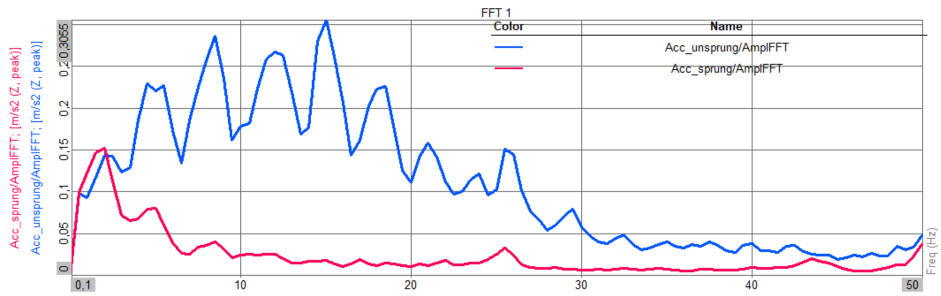

Figure 3.2.2

Note that with the time history reported in Figure (3.2.1) very poor results are obtained. For this reason, the time history is cut to focus only on the wood plank passages and to remove the sections with the engine idling. While this improves the quality of the modal parameters identification it inevitably reduces the length of the recording and, as a consequence, the number of averages (not good). Figure (3.2.2) confirms the low quality of the result.

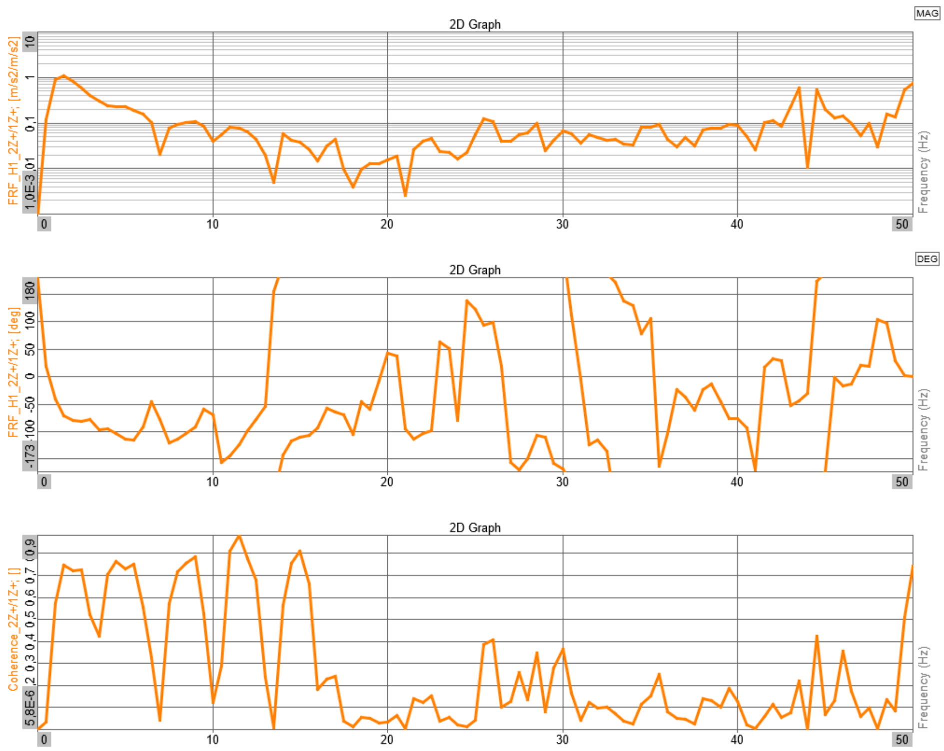

Figure 3.2.3

- one peak in the expected frequency range for the first mode is clearly visible
- coherence is quite high but it is important to remember that the number of averages performed is low and this is partially influencing the result

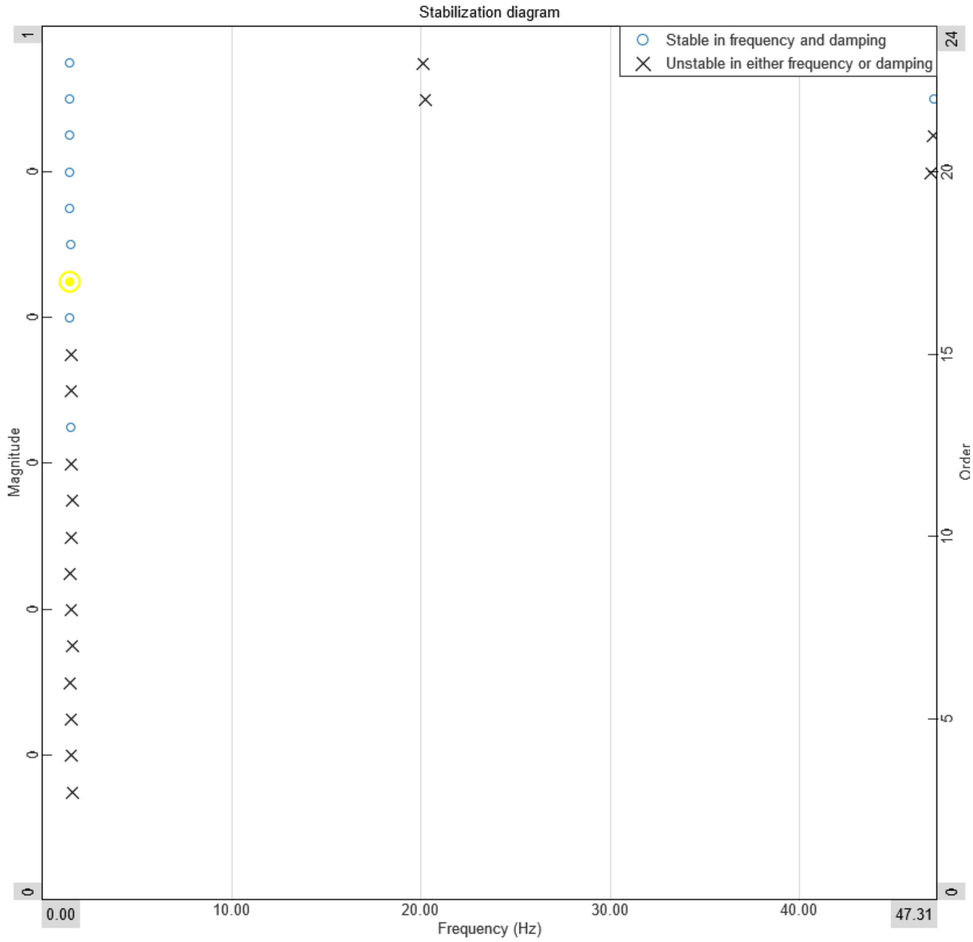

Figure 3.2.4

- the stabilization diagram confirms that the program has confidently identified the first natural frequency of the system. The comparison between the synthetized and measured FRF is reported in Figure (3.2.5) while the value of the natural frequency and the non dimensional damping are reported in Table (1)

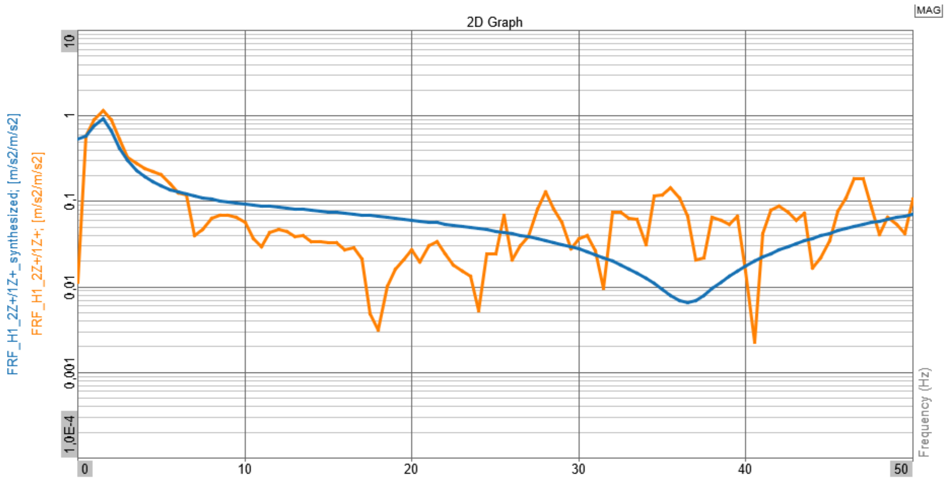

Figure 3.2.5

Quantity	Value
Damped frequency	1.53 Hz
Damping ratio	34.4%

Table 1

3.3 Driving on urban road (undefined path)

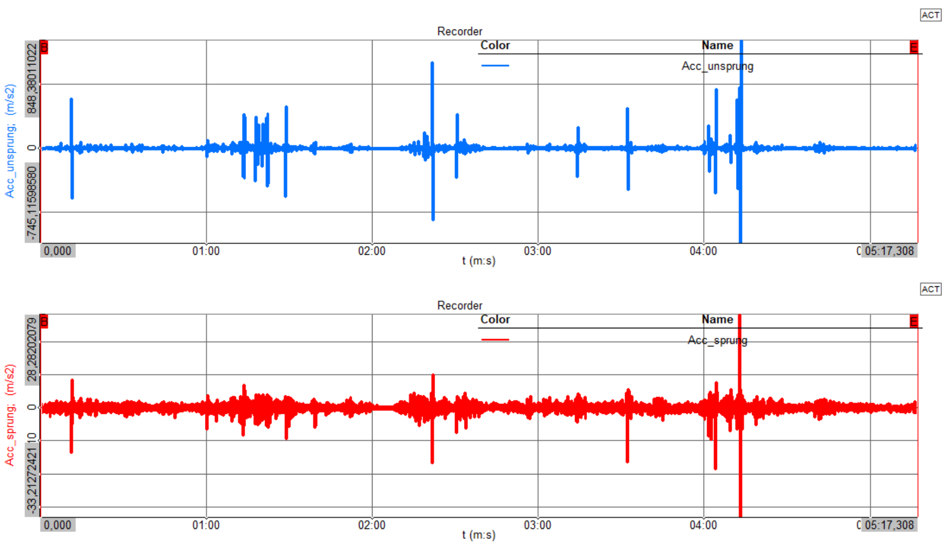

Figure 3.3.1

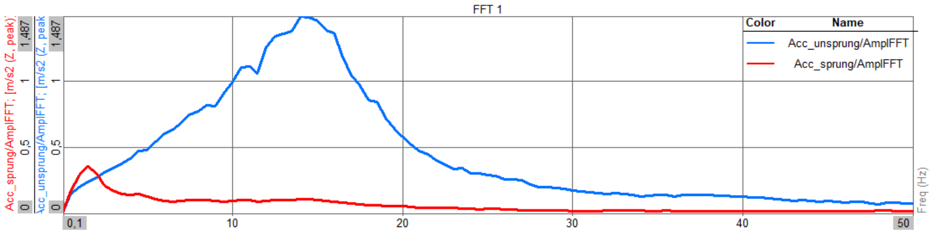

Figure 3.3.2

- two distinct peaks can be identified
  - the first one at very low frequency is associated with the sprung mass heave/bounce oscillation
  - the second one at higher frequency is associated with the unsprung mass oscillation
- this is exactly the behavior expected from a working suspension system, acting as a low pass filter that isolates the chassis from high frequency vibrations

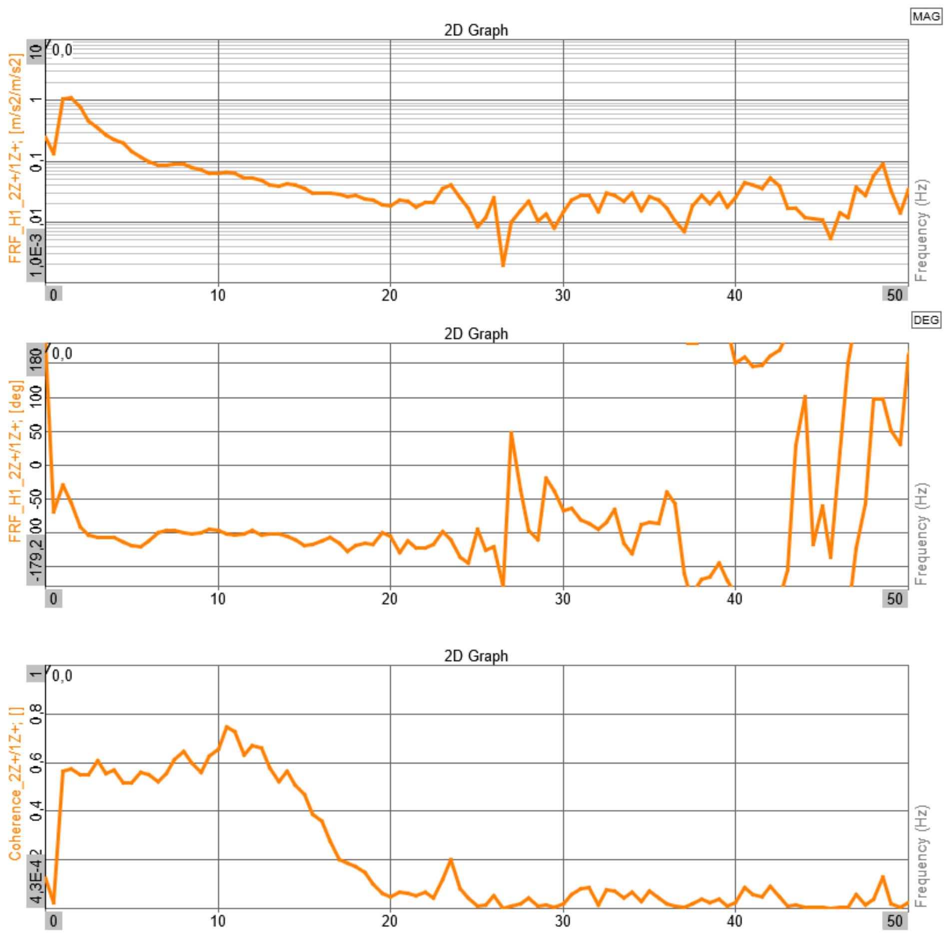

Figure 3.3.3

- a drop before the first peak is clearly visible. Coherence also drops; therefore, it is likely related to measurement noise or to the sprung mass accelerometer lower frequency bound
- one peak is visible in the low frequency range while the second one at higher frequency does not appear. This is clearly related to the behavior of the system described in Figure (3.3.2), where the ratio between the output (red - sprung mass acc.) and input signal (blue - unsprung mass acc) is low. Because the high frequency vibration is so heavily attenuated, a sharp drop in the coherence at frequencies higher than 15 Hz is observed

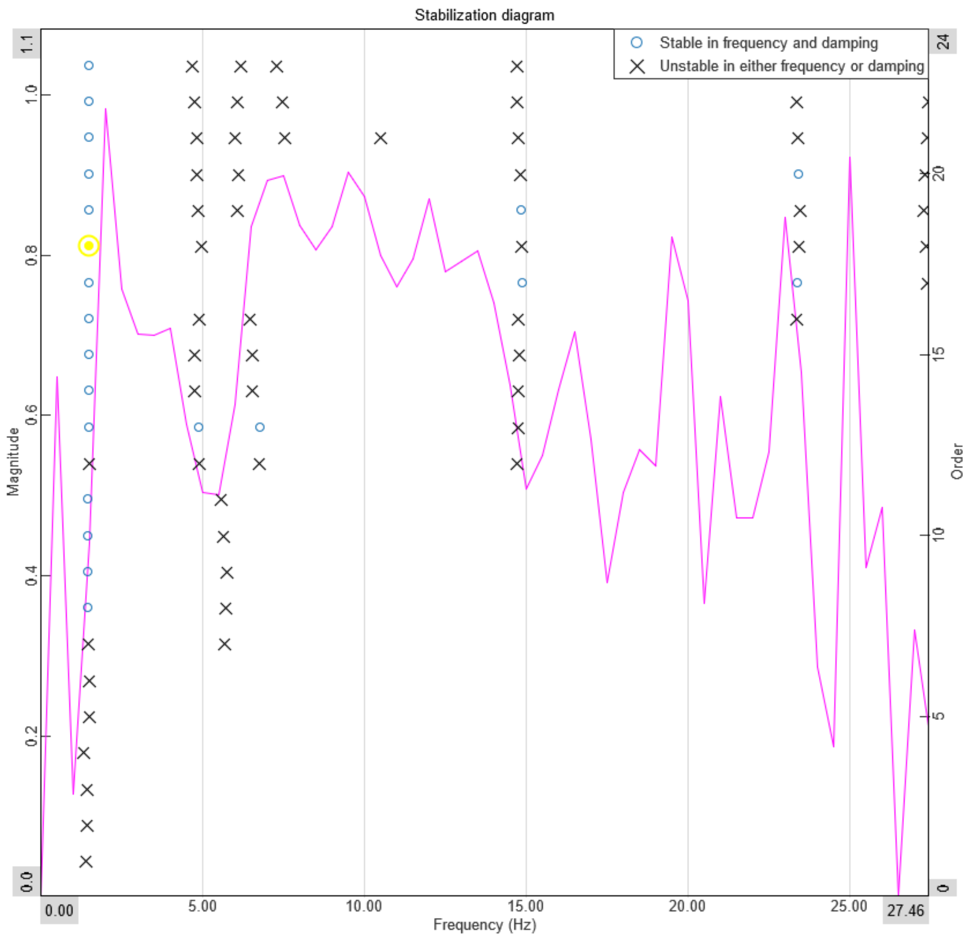

Figure 3.3.4

- the stabilization diagram confirms that the program has confidently identified the first natural frequency of the system (sequence of stable poles and Mode Indicator Function close to 1)
- Table (2) reports the values of the modal parameters computed by Dewesoft X and Figure (3.3.5) reports the visual comparison between the computed and approximated FRF, thus confirming the quality of the results

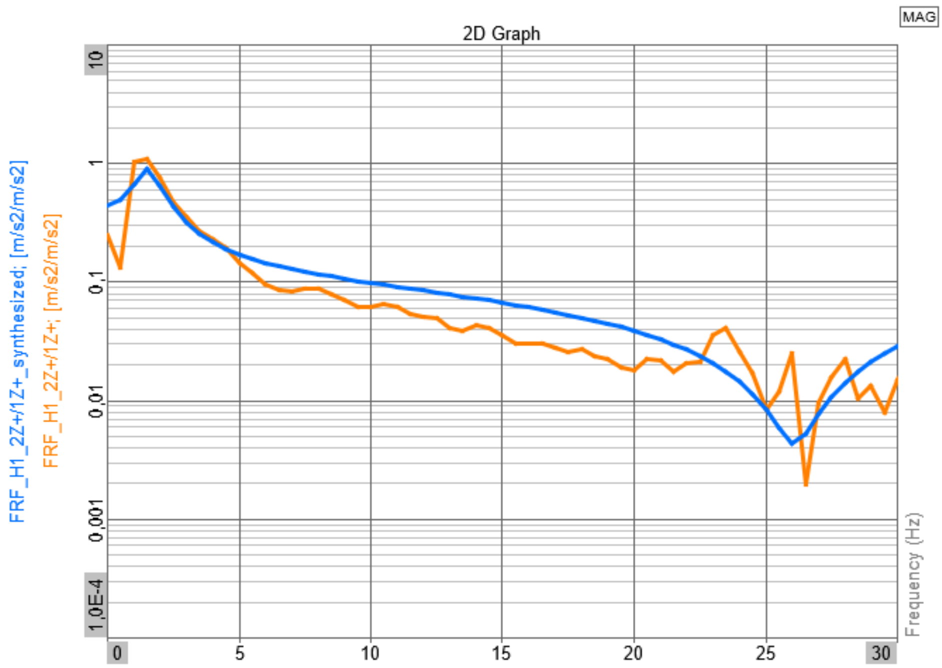

Figure 3.3.5

Quantity	Value
Damped frequency	1.52 Hz
Damping ratio	30.99%

Table 2

3.4 Driving on urban road (well defined path)

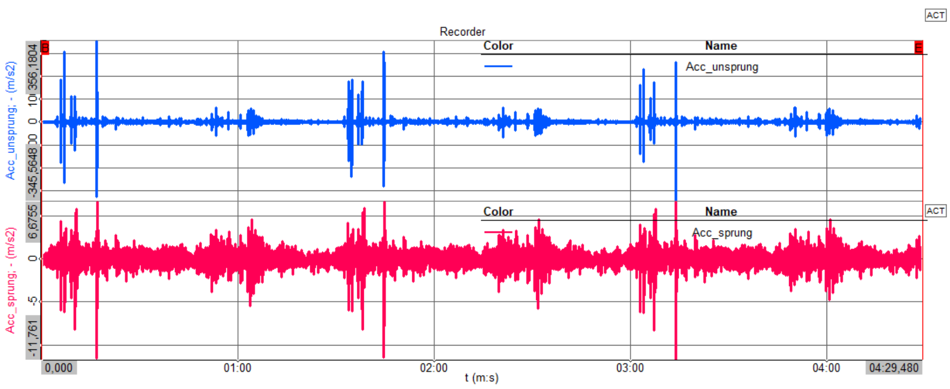

Figure 3.4.1

- the three laps completed around the closed loop path are clearly identified by repeated spike sequences

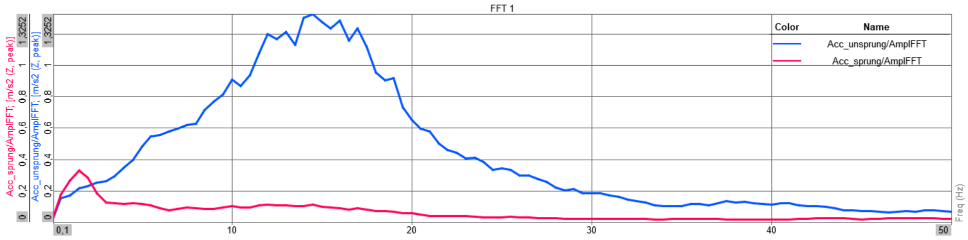

Figure 3.4.2

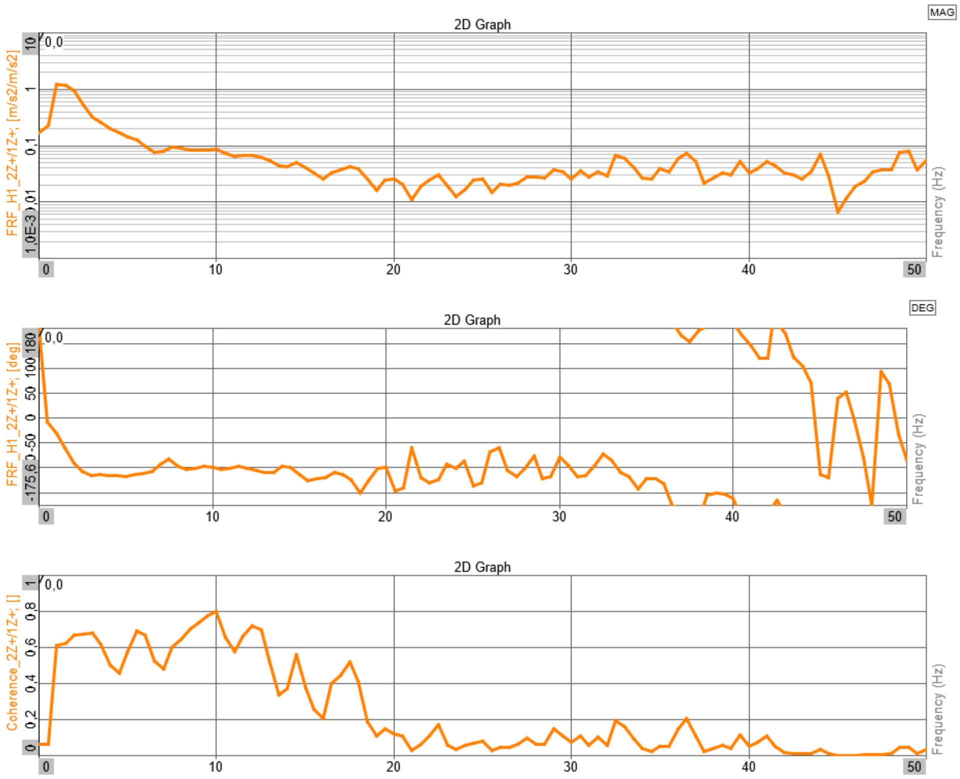

Figure 3.4.3

- the overall shape of the FRF is very similar to the one observed previously, however, the results from the modal analysis module are slightly different

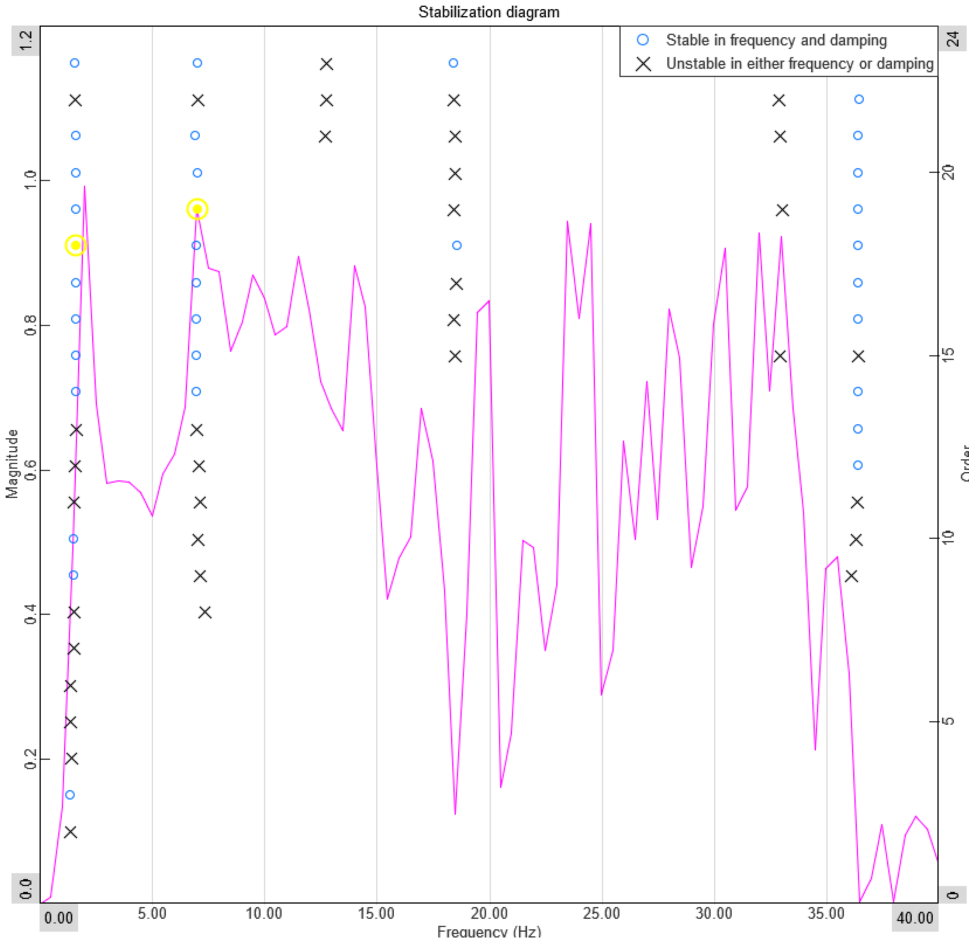
Stabilization diagram
Stable in frequency and damping
Unstable in either frequency or damping
Magnitude
Order
Frequency (Hz)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.00
5.00
10.00
15.00
20.00
25.00
30.00
35.00
40.00
0
5
10
15
20
24

Figure 3.4.4

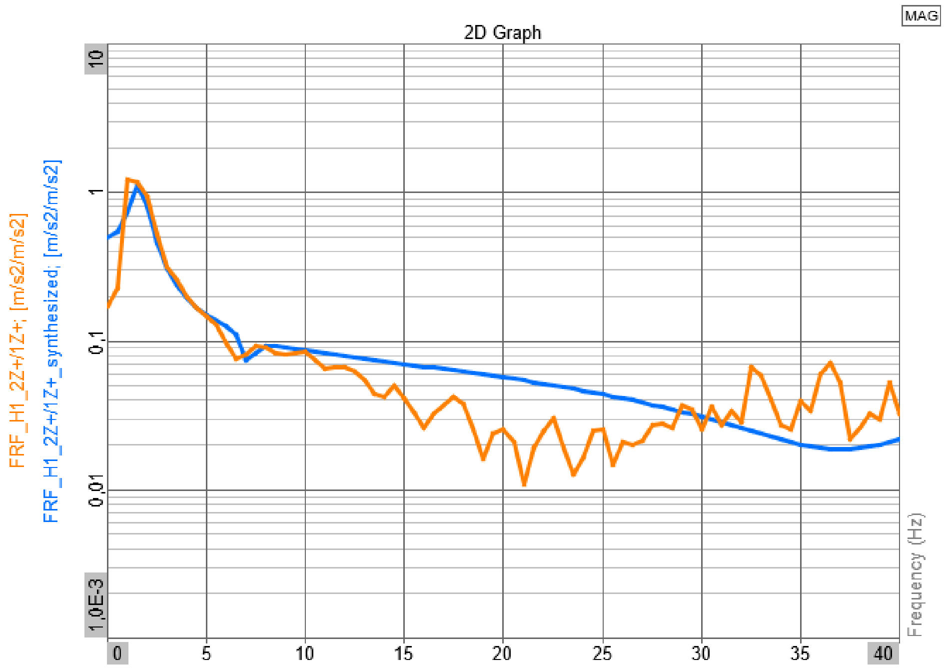

Figure 3.4.5

Quantity	Value
Damped frequency 1	$1.62\,Hz$
Damping ratio 1	25.015%
Damped frequency 2	$7.03\,Hz$
Damping ratio 2	6.40%

Table 3

- the stabilization diagram (Figure (3.4.4)) suggests that the program has confidently identified the first natural frequency of the system (as in the previous cases)
- unexpectedly, the program also identifies a second natural frequency at $\approx 7\,Hz$. Initially, this result might seem contradictory, as passenger vehicles are designed **not** to have vertical suspension modes in the $4\,Hz \div 8\,Hz$ range to avoid severe passenger discomfort

It is necessary to determine if this is a misleading numerical result, an excitation related to the engine rotation (RPM), or a localized structural mode captured only in this specific driving scenario. To deepen the analysis of this unexpected result and determine its physical origin, the spectrogram of the two acceleration signals is taken into consideration (Figure (3.4.6)):

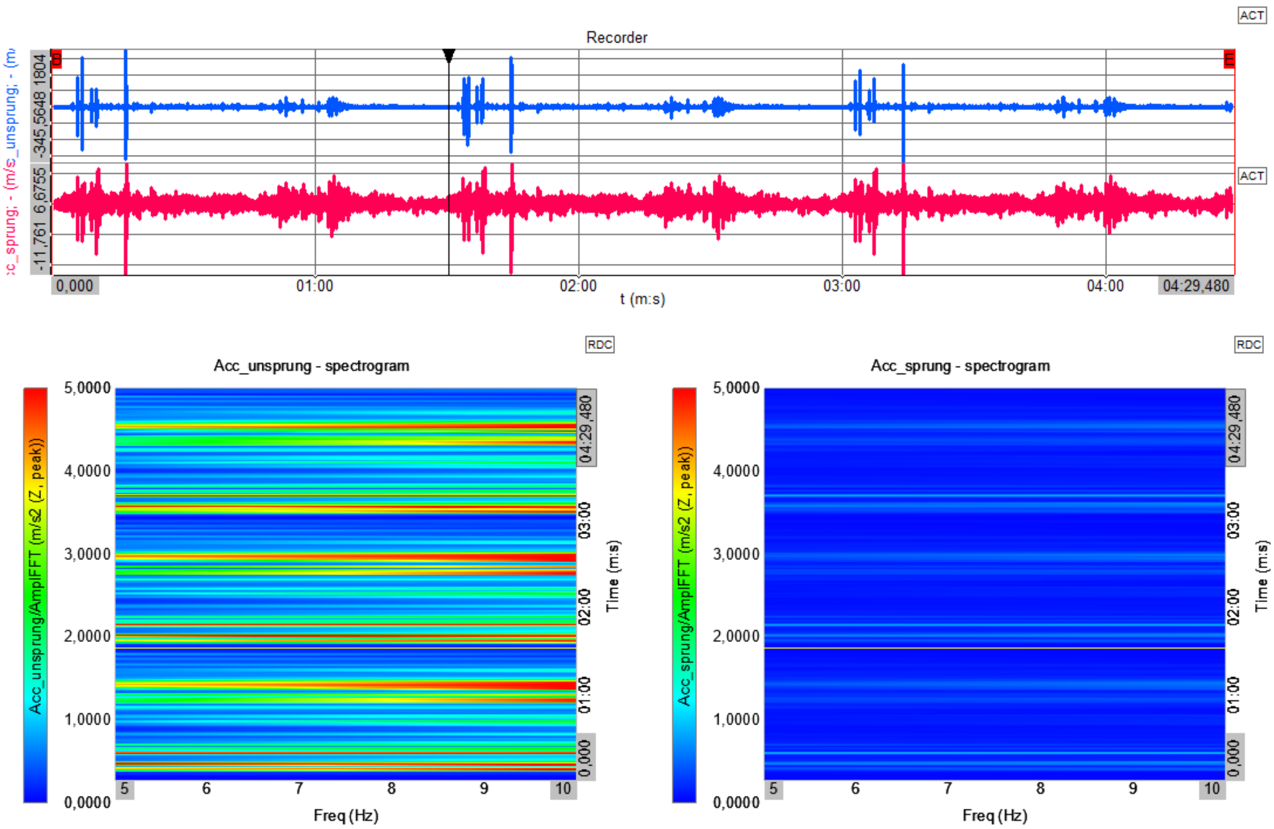

Figure 3.4.6

- the 7 Hz harmonic component is excited mainly in the unsprung mass acceleration signal (left diagram) and specifically in correspondence of road bumps (see the position of the time cursor between minutes 1 and 2)

This impact-related behavior proves that the peak is not related to a continuous excitation, such as engine rotational speed, nor is it a local chassis mode (as it is largely absent in the sprung mass signal). Instead, this harmonic component might be related to a low-frequency vertical vibration (maybe even the actual second natural frequency of the system) from which the sprung mass is intentionally isolated to maintain ride comfort (the small coherence drop in Figure (3.4.3) and the much smaller damping ratio value in Table (3) suggest this).

4 Conclusions

4.1 Results comparison

Driving Scenario	Damped Frequency [Hz]	Damping Ratio [%]
Driving over wood plank	1.53	34.4
Driving on urban road (undefined path)	1.52	30.99
Driving on urban road (well defined path)		
– Mode 1	1.62	25.015
– Mode 2	7.03	6.40

Table 4

Despite the expectations, the results are slightly different from each other but the variability of the first mode is compatible with the large differences between the scenarios and the actual non-linear behavior of the system (the damping coefficient is velocity dependent and bump and rebound behavior is different).

Because the suspension system is designed to isolate the sprung mass from high-frequency wheel vibrations, the ratio between sprung and unsprung mass acceleration at those higher frequencies is inherently low. Therefore, the measured FRF correctly displays only a single peak perfectly illustrating the suspension's function as a mechanical low-pass filter." Increasing the frequency resolution changes the results only slightly but they never match. The values of damping ratio obtained are also reasonable compared to the typical values for road vehicles.

4.2 NAVION unavailability

In section (2.1) it was mentioned that an inertial navigation unit NAVION is also used but it is never mentioned or used in the analysis of the results. The reason for this is that a severe malfunctioning was recorded and none of the channels were available except roll and pitch angles. Considering that these two outputs are not measured directly but are computed starting from INS measured data, their values are not considered valid.

4.3 Possible improvements

A few options that might be taken into consideration as further developments/improvements for this experiment are reported below:

1. perform the test with the engine off and pushing the van
    - advantages (✓):

- engine related vibration (noise) is completely removed

- disadvantages (×):
  - extremely low speed
  - the weight of the van is considerable, this option may be considered in case a lighter vehicle is used

2. change the position of the sprung mass accelerometer and place it closer to the dome of the damper or alternatively on a rear wheel (in general, place it closer to where the damper transmits loads to the chassis or farther from the engine)

   - advantages (✓):
     - engine related vibration (noise) is reduced
     - the effect of the chassis structural damping and dynamics are reduced
   - disadvantages (×):
     - according to the type of vehicle, accessing the dome of the damper and securing the accelerometer might be hard

3. jump on the van instead of driving

   - advantages (✓):
     - engine related vibration (noise) is completely removed
     - higher repeatability

4.4 Final remarks

Ultimately, the empirical data acquired during the experimental scenarios successfully validated the theoretical behavior described by the 2DOF quarter-car model. The frequency spectrum analysis clearly demonstrates the suspension system acting as a low-pass filter: a distinct low-frequency peak associated with the sprung mass heave and a higher-frequency peak corresponding to the unsprung mass oscillation are (almost) consistently identified.

In conclusion, despite the low signal-to-noise ratio encountered with the sprung mass accelerometer and the hardware unavailability of the NAVION inertial platform, the experimental setup managed to characterize the fundamental vertical dynamics of the subject vehicle. The acquired data provides valuable real-world insights that bridge the gap between theoretical vehicle dynamics models and practical physical testing.

Special thanks to the Dewesoft team in Vitinia for the huge help with the experiment setup and the support throughout the activity.

Figure 4.4.1